

StudyHub

Project Documentation Report

A MERN-based Educational Platform
for Notes Sharing and Student Collaboration

Submitted by:
[Ishwar Kawade](#)

November 7, 2025

Table of Contents

1 Introduction	4
1.1 Existing System and Need for System	4
1.2 Scope of System	4
1.3 Operating Environment – Hardware and Software	4
2 Proposed System	5
2.1 Feasibility Study	5
2.2 Objectives of Proposed System	5
2.3 Users of System	5
3 Analysis and Design	6
3.1 System Requirements (Functional and Non-Functional)	6
3.2 Module Hierarchy Diagram	6-7
3.3 Sample Input and Output Screens	7-8
4 Coding	9
4.1 Code Snippets	9
5 Limitations of Proposed System	10
6 Proposed Enhancements	10
7 Conclusion	10

1. Introduction

StudyHub is a comprehensive MERN (MongoDB, Express.js, React, Node.js) stack web application designed to facilitate educational content sharing and collaboration among students and teachers. The platform enables seamless sharing of study materials, real-time communication, and community-driven learning.

1.1 Existing System and Need for System

Traditional educational systems face several challenges:

- Limited accessibility to quality study materials
- Lack of centralized platforms for note sharing
- Inefficient communication between students and teachers
- No organized community-driven learning environment
- Difficulty in finding subject-specific resources

StudyHub addresses these challenges by providing a unified platform for educational resource sharing and collaboration.

1.2 Scope of System

The StudyHub system encompasses the following features:

- User authentication and role-based access (Students and Teachers)
- Upload and download study materials (notes, PDFs, documents)
- Community-based discussion forums
- Real-time messaging using Socket.IO
- Poll creation and voting system
- Notification system for community activities
- Teacher dashboard for content management
- User profile management
- Search and filter functionality for notes

1.3 Operating Environment – Hardware and Software

Hardware Requirements:

- Processor: Intel Core i3 or higher
- RAM: Minimum 4GB (8GB recommended)
- Storage: Minimum 500MB free space
- Internet connection: Broadband (minimum 2 Mbps)

Software Requirements:

- Operating System: Windows 10/11, macOS, or Linux
- Node.js: v14.0.0 or higher
- MongoDB: v4.4 or higher
- Web Browser: Chrome, Firefox, Safari, or Edge (latest versions)
- Code Editor: VS Code, Sublime Text, or similar

2. Proposed System

StudyHub is proposed as a modern, scalable, and user-friendly educational platform that leverages the MERN stack to provide a seamless experience for students and teachers.

2.1 Feasibility Study

Technical Feasibility:

- MERN stack provides robust and proven technologies
- Cloud deployment ensures scalability and accessibility
- Real-time features achievable with Socket.IO
- Cloudinary integration for efficient file management

Economic Feasibility:

- Open-source technologies reduce development costs
- Cloud hosting provides cost-effective scaling
- Minimal maintenance overhead with modern frameworks

Operational Feasibility:

- Intuitive user interface ensures easy adoption
- Responsive design works across all devices
- Minimal training required for end users

2.2 Objectives of Proposed System

- Create a centralized platform for educational content sharing
- Enable real-time collaboration between students and teachers
- Provide organized community forums for subject-specific discussions
- Facilitate easy upload, search, and download of study materials
- Implement secure authentication and authorization
- Deliver responsive and accessible user experience
- Enable teachers to manage and share educational content

2.3 Users of System

Primary Users:

- Students: Upload/download notes, participate in communities, chat with peers
- Teachers: Share educational content, manage notes, interact with students

Secondary Users:

- Educational Institutions: Monitor and promote academic collaboration
- Administrators: Manage platform content and user activities

3. Analysis and Design

3.1 System Requirements

Functional Requirements:

- User Registration and Authentication (JWT-based)
- Role-based access control (Student/Teacher)
- Upload notes with metadata (subject, semester, branch)
- Search and filter notes by various parameters
- Download notes in multiple formats
- Create and join community forums
- Real-time messaging within communities
- Create polls and vote on community topics
- Notification system for updates
- User profile management with statistics

Non-Functional Requirements:

- Performance: Page load time < 3 seconds
- Scalability: Support for 10,000+ concurrent users
- Security: Encrypted passwords, secure token management
- Reliability: 99.5% uptime
- Usability: Intuitive UI with minimal learning curve
- Compatibility: Cross-browser and cross-device support
- Maintainability: Modular code architecture

3.2 Module Hierarchy Diagram

The system is organized into the following modules:

Frontend Modules:

- Authentication Module (Login, Signup, Teacher Signup)
- Home Module (Landing page, Features showcase)
- Notes Module (Upload, Browse, Search, Download)
- Communities Module (Browse, Join, Create, Discussion)
- Profile Module (User stats, Settings, Activity)
- Teacher Dashboard Module (Content management)

Backend Modules:

- Authentication Service (JWT, Bcrypt)
- User Management Service
- Notes Management Service (Upload, Delete, Retrieve)
- Community Service (CRUD operations)
- Messaging Service (Socket.IO)
- Notification Service
- Poll Service
- Teacher Service

Database Schema:

- Users Collection (authentication, profile data)
- Notes Collection (study materials metadata)
- Communities Collection (forum data)
- Messages Collection (chat history)
- Polls Collection (voting data)
- Notifications Collection (user alerts)

3.3 Sample Input and Output Screens

Note: Screenshots of the application would be inserted here showing:

- Home Page - Landing page with features and call-to-action
- Login/Signup Page - User authentication interface
- Notes Page - Browse and search notes with filters
- Upload Page - Form for uploading study materials
- Communities Page - List of available communities
- Community Detail Page - Discussion forum with real-time chat
- Profile Page - User statistics and uploaded content
- Teacher Dashboard - Content management interface

[Screenshots would be placed in this section showing the actual UI of the application]

4. Coding

StudyHub is built using modern web technologies with clean, maintainable code architecture.

4.1 Code Snippets

Technology Stack:

- Frontend: React 19.1.0, React Router, Material-UI, TailwindCSS
- Backend: Node.js, Express 5.1.0, MongoDB, Mongoose
- Real-time: Socket.IO 4.8.1
- Authentication: JWT, Bcrypt
- File Storage: Cloudinary
- Additional: Compression, CORS, Multer, Nodemailer

Key Code Implementations:

Server Configuration (server.js):

```
import express from 'express';
import mongoose from 'mongoose';
import socketIO from 'socket.io';
import compression from 'compression';

const app = express();
app.use(compression());
app.use(express.json({ limit: '50mb' }));

// Routes
app.use('/api/auth', authRoutes);
app.use('/api/notes', notesRoutes);
app.use('/api/communities', communityRoutes);
```

Frontend Routing (App.jsx):

```
import { BrowserRouter, Routes, Route } from 'react-router-dom';

function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/notes" element={<Notes />} />
        <Route path="/communities" element={<Communities />} />
      </Routes>
    </BrowserRouter>
  );
}
```


5. Limitations of Proposed System

- Limited to text-based notes and PDFs (no video support currently)
- Requires stable internet connection for real-time features
- File upload size restricted to 50MB
- Search functionality limited to metadata (not full-text search)
- No offline access to downloaded notes
- Community moderation features are basic
- No built-in plagiarism detection for uploaded content

6. Proposed Enhancements

- Video lecture upload and streaming capabilities
- AI-powered content recommendation system
- Advanced search with full-text indexing
- Mobile applications (iOS and Android)
- Offline mode with progressive web app (PWA)
- Integration with popular LMS platforms
- Advanced analytics and progress tracking
- Gamification features (badges, leaderboards)
- Multi-language support
- Content versioning and collaboration features
- Integration with citation management tools
- Automated content quality assessment

7. Conclusion

StudyHub successfully addresses the need for a centralized, collaborative educational platform. By leveraging modern web technologies (MERN stack), the system provides:

- Seamless user experience across devices
- Real-time collaboration features
- Scalable architecture for future growth
- Secure and efficient content management

The platform has demonstrated its capability to facilitate knowledge sharing among students and teachers, creating a vibrant learning community. With planned enhancements, StudyHub has the potential to become a comprehensive educational ecosystem.

The project successfully meets its core objectives of providing an accessible, user-friendly platform for educational content sharing while maintaining high standards of code quality, security, and performance.

Project URL: <https://studyhub4all.netlify.app>

Repository: [StudyHub GitHub Repository](#)

--- End of Report ---